

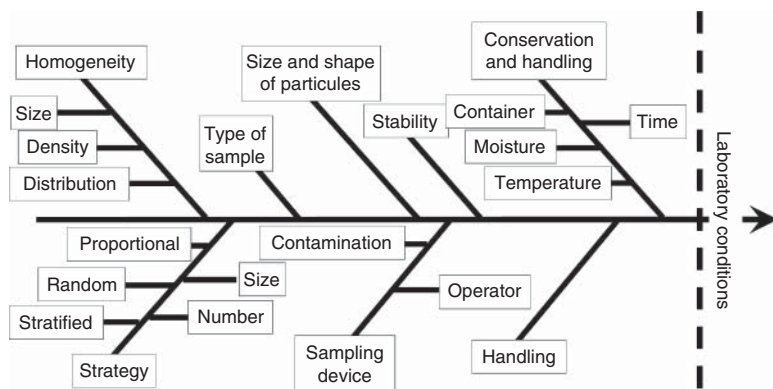


Figure 8.7 Cause to effect diagram of the sampling operation.

living materials, such as foods or biological tissues and fluids. In this case, the underlying assumptions concerning the sampling constant or the mode of aggregation of the analyte are not easy to verify or even fulfilled. It involves a significant analytical cost that is not always justified [11].

According to the general GUM procedure for MU estimation described in Section 6.2, the first step consists of listing the different sources of uncertainty specific to the sampling and constructing a cause-to-effect diagram, like Figure 7.2. The diagram adapted to sampling is reproduced in Figure 8.7. It implies that studying each component would require a daunting number of measurements.

According to the GUM general procedure, the sources of uncertainty can be grouped. In Section 7.2, it is stated that global analytical MU can generally be expressed as the combination of three main standard uncertainties, namely:

The analytical process (analytical uncertainty)	$u(Z_m)$
The sampling process (sampling uncertainty)	$u(Z_s)$
The measurand definition (definitional uncertainty)	$u(Z_d)$

From this statement, the combined standard variance of any analytical measurement can be described by Equation (8.6):

Combined standard variance

$$u_c^2(Z) = u^2(Z_m) + u^2(Z_s) + u^2(Z_d) \quad (8.6)$$

Leaving aside the definitional uncertainty, which depends very much on the traceability of the measurand to the SI unit system as explained in Section 4.1.3, the standard variance of a result can be written in a simplified form:

The simplified form of the standard variance

$$u^2(Z) = u^2(Z_m) + u^2(Z_s) \quad (8.7)$$

If the part due to the analytical process $u(Z_m)$ has been estimated beforehand, which is the case when the MAP and uncertainty function has already been established, it is easy to extract the sampling uncertainty:

$$u^2(Z_s) = u^2(Z) - u^2(Z_m)$$

$$u(Z_s) = \sqrt{u^2(Z) - u^2(Z_m)}$$

Recently, several procedures adapted to the analytical sciences have been published. These propositions are diverse and often require sophisticated means to collect the data and perform the calculation [8, 13].

8.3.2 Procedure of Homogeneity Check

In the assessment of sampling uncertainty, the estimation of homogeneity is one of the key issues. Unfortunately, the deterministic modeling described above is not universal, but another can be proposed. It is rather easy as it is based on statistical tools already developed in Chapters 3 and 5, such as ANOVA and tolerance intervals. The starting point is Annex B of the ISO 13528 standard which describes an experimental design for evaluating the homogeneity of the test materials used in proficiency testing schemes (PTS).

This Annex is entitled “Verification of homogeneity and stability of proficiency test entities.” It describes an operating procedure for the verification of homogeneity that can be adapted as follows:

- From the lot used for the study, select N samples, i.e. preferably called sampling units, with $N \geq 10$.
- For each sample, prepare two test portions using appropriate techniques to minimize differences between them.
- Analyze the $2 \times N$ test portions in random order but under repeatability conditions.
- Applying a one-way random effect ANOVA, compute the grand mean $\bar{\bar{Z}}$, within-sample standard deviation s_r and between-samples standard deviation s_B as described in Section 3.2.

To decide if the homogeneity of the lot is acceptable, the standard ISO 13528 proposes to compare the inter-sample standard deviation s_B to a reference standard deviation σ called “standard deviation for proficiency assessment,” which is a “measure of dispersion used in the evaluation of results of proficiency testing, based on the available information” and is supposed to allow an assessment of laboratory proficiency. The acceptance criterion used to decide if the lot is sufficiently homogeneous is:

Acceptance criterion

$$\frac{s_B}{\sigma} \leq 0.3$$

Squared criterion

$$\frac{s_B^2}{\sigma^2} \leq 0.1$$

The squared acceptance criterion can be interpreted as follows: the between-samples variance must not exceed 10% of the reference variance. In practice, the proposed acceptance criterion seemed rather strict, and in a more recent standard applied to microbiology under revision (ISO 22117), it was set at 0.5 so that the between-samples variance must not exceed 25% of the reference variance. To adapt this procedure to sampling uncertainty, a simple suggestion is to replace this ratio with the variance ratio A already described (Eq. 5.5, page 126). The acceptance criterion becomes:

Possible acceptance criteria

$$A = \frac{s_B^2}{s_r^2} \leq 0.1 \text{ or } 0.25 \quad (8.8)$$

In numerous accuracy profile examples shown in Chapters 5 and 7, the variance ratio A is much higher than one of these acceptance limits. However, these data are collected on purpose under intermediate precision conditions and not repeatability. Therefore, it is mandatory that the measurements made to check the homogeneity be obtained under repeatability conditions to avoid any additional source of variation being interpreted as heterogeneity. The standard variance $u^2(Z)$ is described by Equation (8.7), and the decision whether it is necessary to incorporate the sampling uncertainty into combined uncertainty can be made on this acceptance criterion. If the A ratio is higher than these limits of 0.1 or 0.25, sampling uncertainty is significant. The second proposal is to calculate the parameters of the β -expectation tolerance interval (β -ETI) from the data collected according to the experimental design described in Annex B of ISO 13528 standard, as explained in Section 7.2. The standard deviation of this interval denoted s_{IT} can be directly interpreted as an estimate of the sampling uncertainty $u(Z_s)$.

8.3.3 Example of Copper in Wheat Flour

Within the framework of the official control of pesticides in wheat flour, the analysis of copper was used to check the homogeneity of a T65 wheat flour lot. Twenty samples were taken from the batch of wheat flour at different randomly selected locations for this check. No prior homogenization was applied to be consistent with the future conditions of use. From each sample, two test portions were analyzed by flame atomic absorption spectrophotometry. This method was selected as it is fast and inexpensive with a well-established analytical uncertainty. Table 8.1 is a compilation of 40 measures performed on separate test portions issued from 20 control samples denoted S01 to S20.

The data are input into the Resource H worksheet after increasing the number of series from 6 to 10. The best way is to add rows in the middle of an existing range so that all formulas are automatically updated, except in columns D and E, where formulas must be manually copied. From the computed parameters, several conclusions can be drawn. Table 8.2 shows the β -ETI results using the notations of this book. When verifying the acceptance criteria proposed by formula (8.8), it is possible to conclude that the batch is heterogeneous because the variance ratio A is much above 0.1 or 0.25. Sample homogeneity needs to be evaluated.

Table 8.1 COPPER – measurements in T65 wheat flour (mg/kg)^{a)}.

Control	Measure 1	Measure 2	Control	Measure 1	Measure 2
S01	4.2	4.3	S11	4.3	4.3
S02	4.5	4.3	S12	4.2	4.2
S03	4.1	4.5	S13	4.5	4.3
S04	4.1	4.2	S14	4.3	4.5
S05	4.1	4.3	S15	4.2	4.4
S06	4.2	4.2	S16	4.2	4.3
S07	4.3	4.1	S17	4.2	4.1
S08	4.1	4.5	S18	4.1	4.3
S09	4.6	4.7	S19	4.5	4.5
S10	4.5	4.6	S20	4.6	4.5

a) Unpublished personal data.

Table 8.2 COPPER – statistical homogeneity parameters.

Parameters	Symbol	Value	Standard deviation	Variance
Average content (mg/kg)	$\bar{\bar{Z}}$	4.316		
Within sample std. dev.	s_r		0.1291	0.0167
Between-samples std. dev.	s_B		0.1175	0.0138
Intermediate precision std. dev.	s_{IP}		0.1746	0.0305
β -ETI standard deviation	s_{TI}		0.1777	0.0316
Effective number of measures	N_E	31.7		
Variance ratio	A	0.828		

Figure 8.8 illustrates this heterogeneity. At this level of concentration, the relative uncertainty of the copper determination in wheat flour is 3.2% giving a coverage interval of [4.18, 4.15] mg/kg. This interval is shown in Figure 8.8 as two dashed lines around the average marked by a thick solid line and underlines the heterogeneity.

It was already demonstrated in Section 7.2.4 and Equation (7.5) and following, that the standard variance of the β -ETI s_{TI}^2 can be regarded as a proper estimation of standard variance $u^2(Z)$. This property is used to give an estimate of the left part of Equation (8.7). On the other hand, the uncertainty function obtained from the copper data gives 3.2% for the relative uncertainty of a measurement of 4.31 mg/kg. This value will allow us to calculate the analytical standard uncertainty $u^2(Z_m)$ without forgetting the coverage factor of 2. The whole calculation is summarized below:

Source	Symbol	Value	Formula
Average content	$\bar{\bar{Z}}$	4.316	
Analytical process	$UR \% (Z_m)$	3.20%	$0.1069 = \frac{(4.316 \times 0.032)}{2}$
	$u(Z_m)$	0.06906	
	$u^2(Z_m)$	0.00477	0.06906×0.06906
Simplified model	$u^2(Z) = s_{TT}^2$	0.03158	Equation (8.7)
Sampling	$u^2(Z_s)$	0.02681	$u^2(Z_s) = u^2(Z) - u^2(Z_m)$
	$u(Z_s)$	0.16373	$u(Z_s) = \sqrt{u^2(Z) - u^2(Z_m)}$
	$UR \% (Z_s)$	7.59%	$UR \% (Z_s) = 0.164/4.316$

In conclusion, for this particular case, the sampling uncertainty represents a significant contribution to the total combined uncertainty. Its relative contribution is about 7.59%, and is higher than the relative uncertainty due to the analytical process, i.e. 3.20%. The models developed for geochemical measurements, and briefly presented at the beginning of this chapter, introduced the concept that sampling uncertainty depends on a sampling constant which is an invariant, independent of the concentration level. This would mean that the 7.59% value can be used for any copper content but also analytes other than copper. However, this statement needs to be verified.

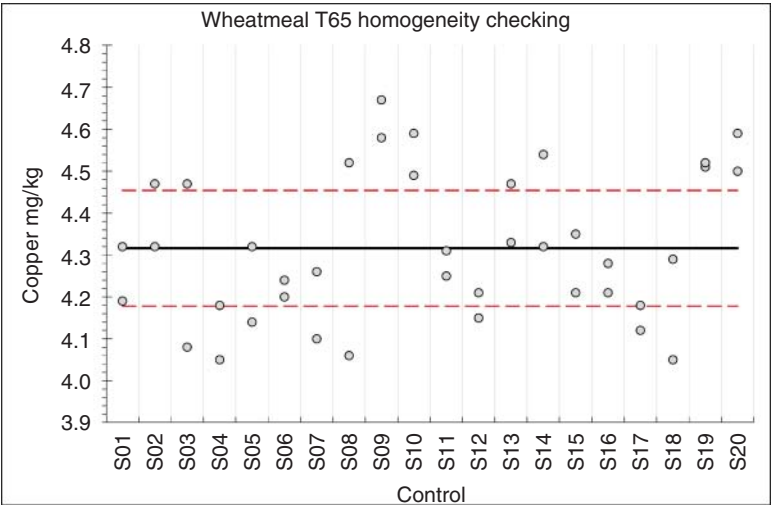


Figure 8.8 COPPER – distribution of measurements and control number.